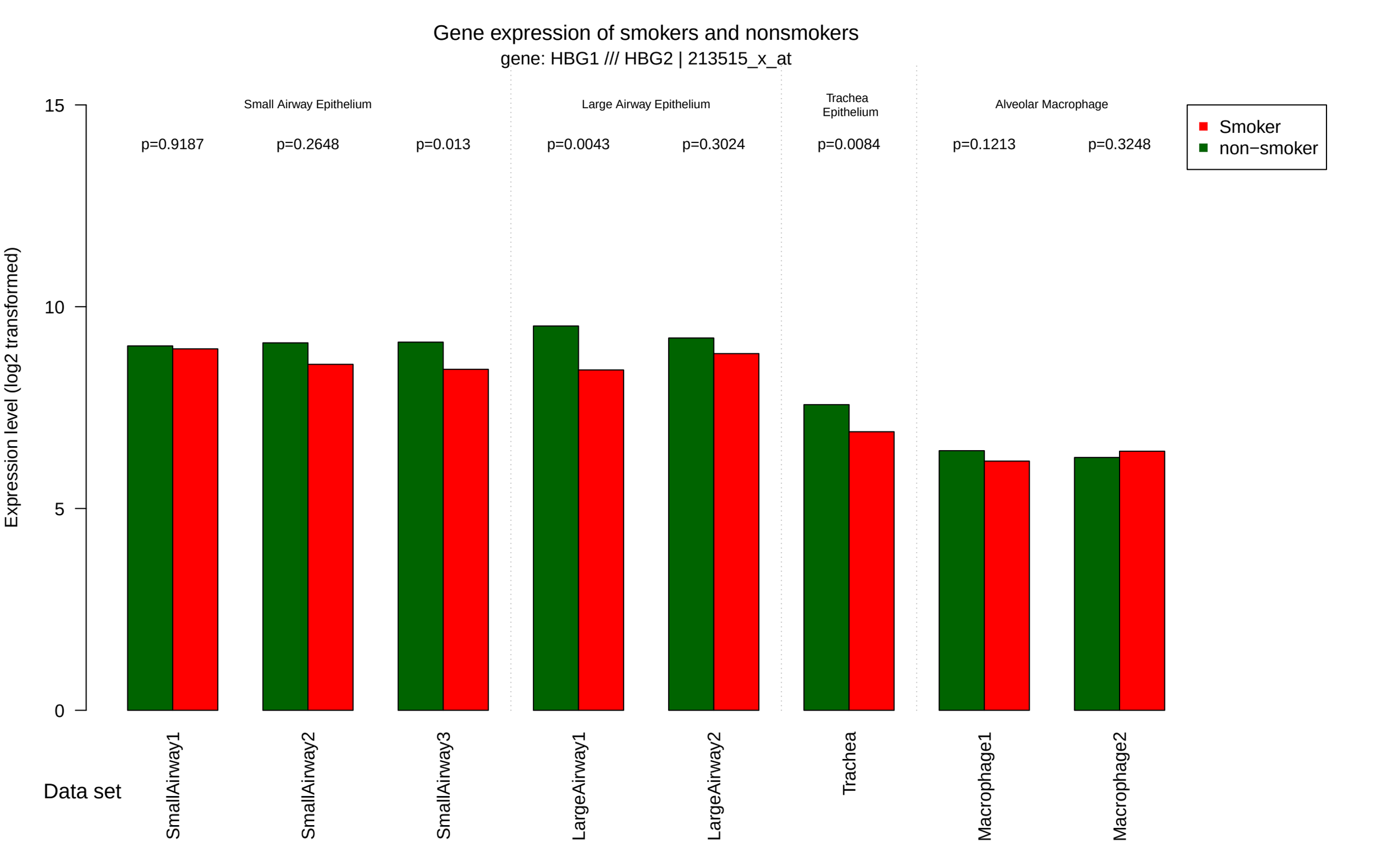
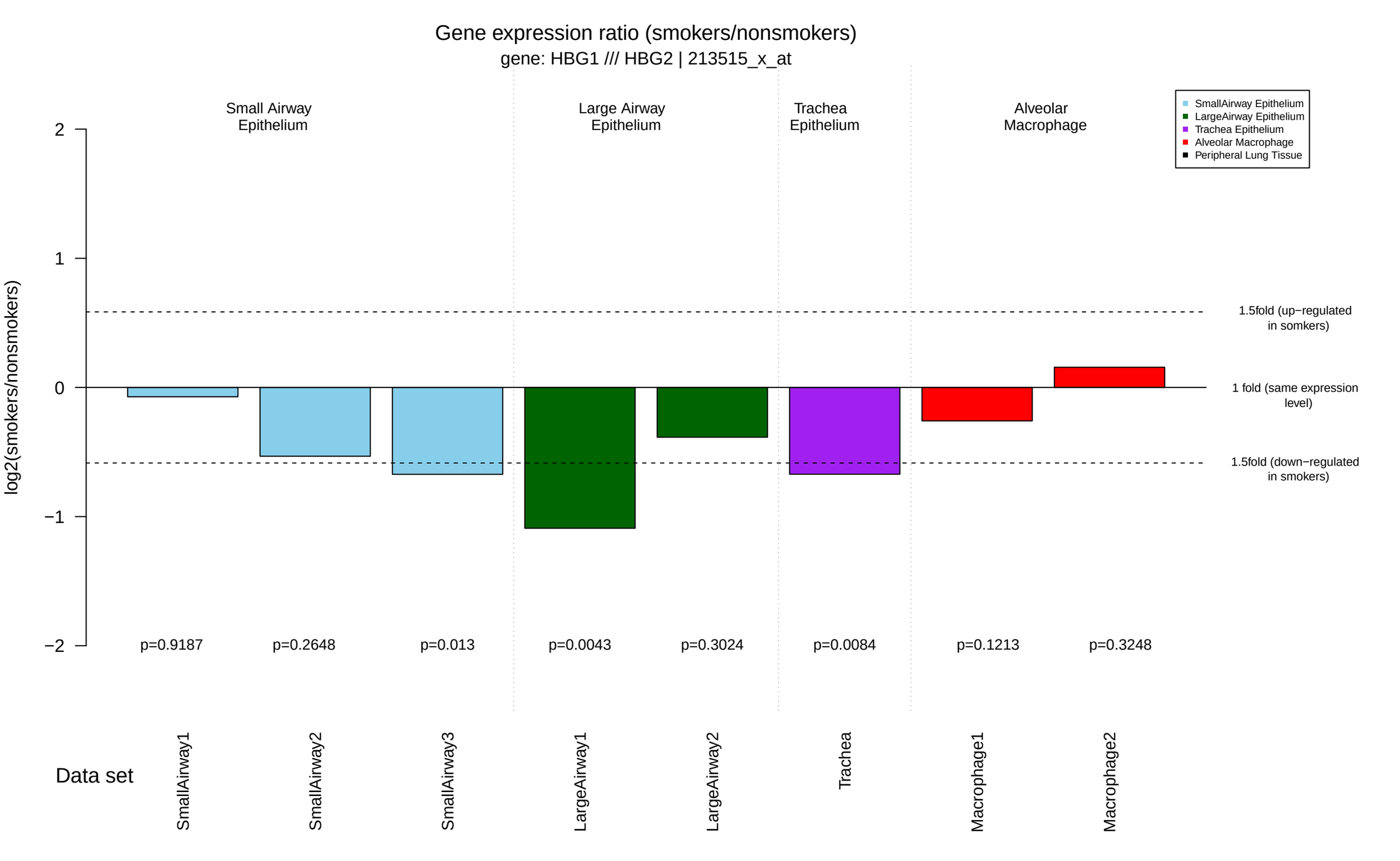
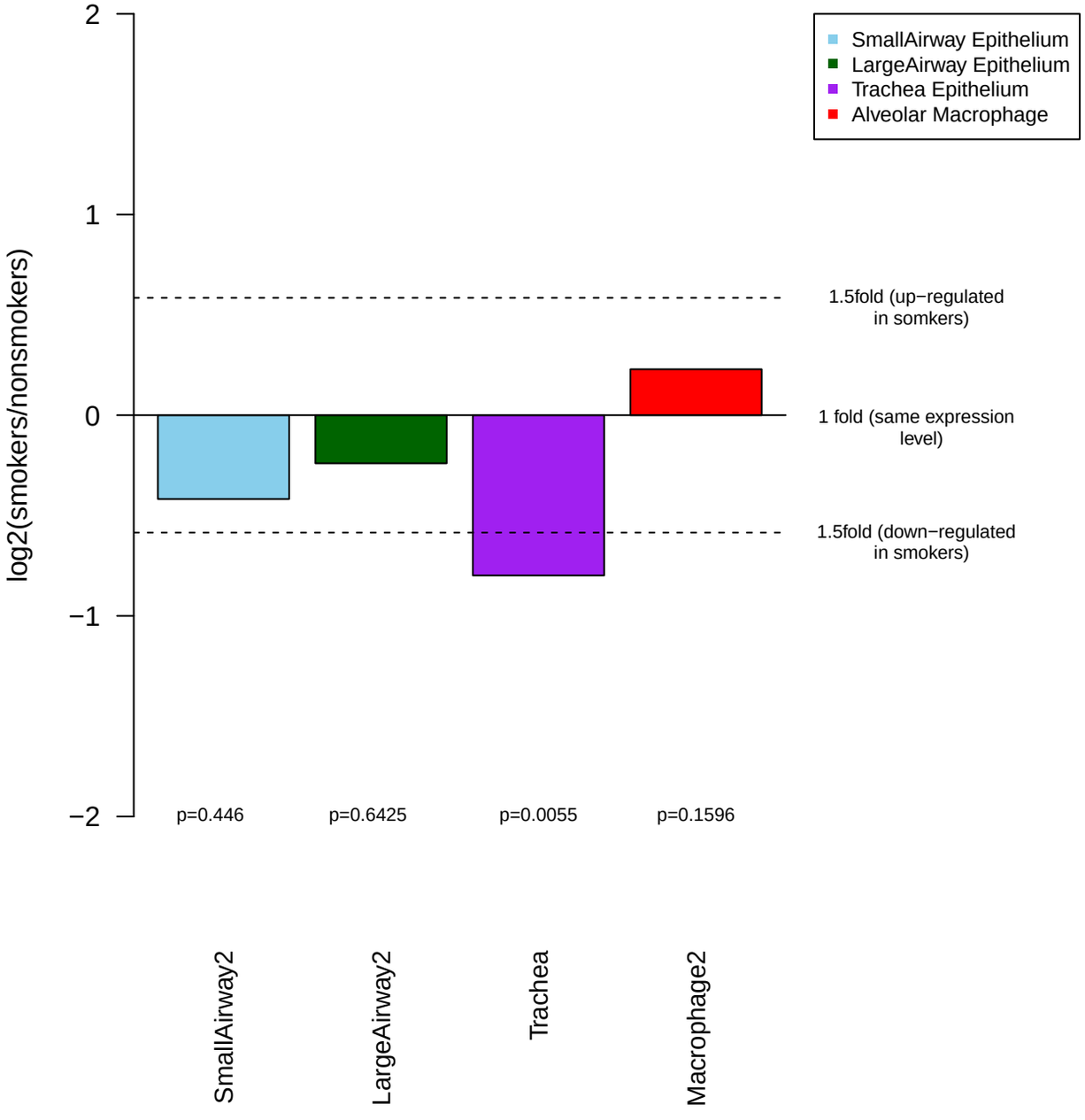

Gene expression of smokers and nonsmokers
gene: HBG1 /// HBG2 | 213515_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9187
p=0.2648
p=0.013
p=0.0043
p=0.3024
p=0.0084
p=0.1213
p=0.3248
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio (smokers/nonsmokers)
gene: HBG1 /// HBG2 | 213515_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9187
p=0.2648
p=0.013
p=0.0043
p=0.3024
p=0.0084
p=0.1213
p=0.3248
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

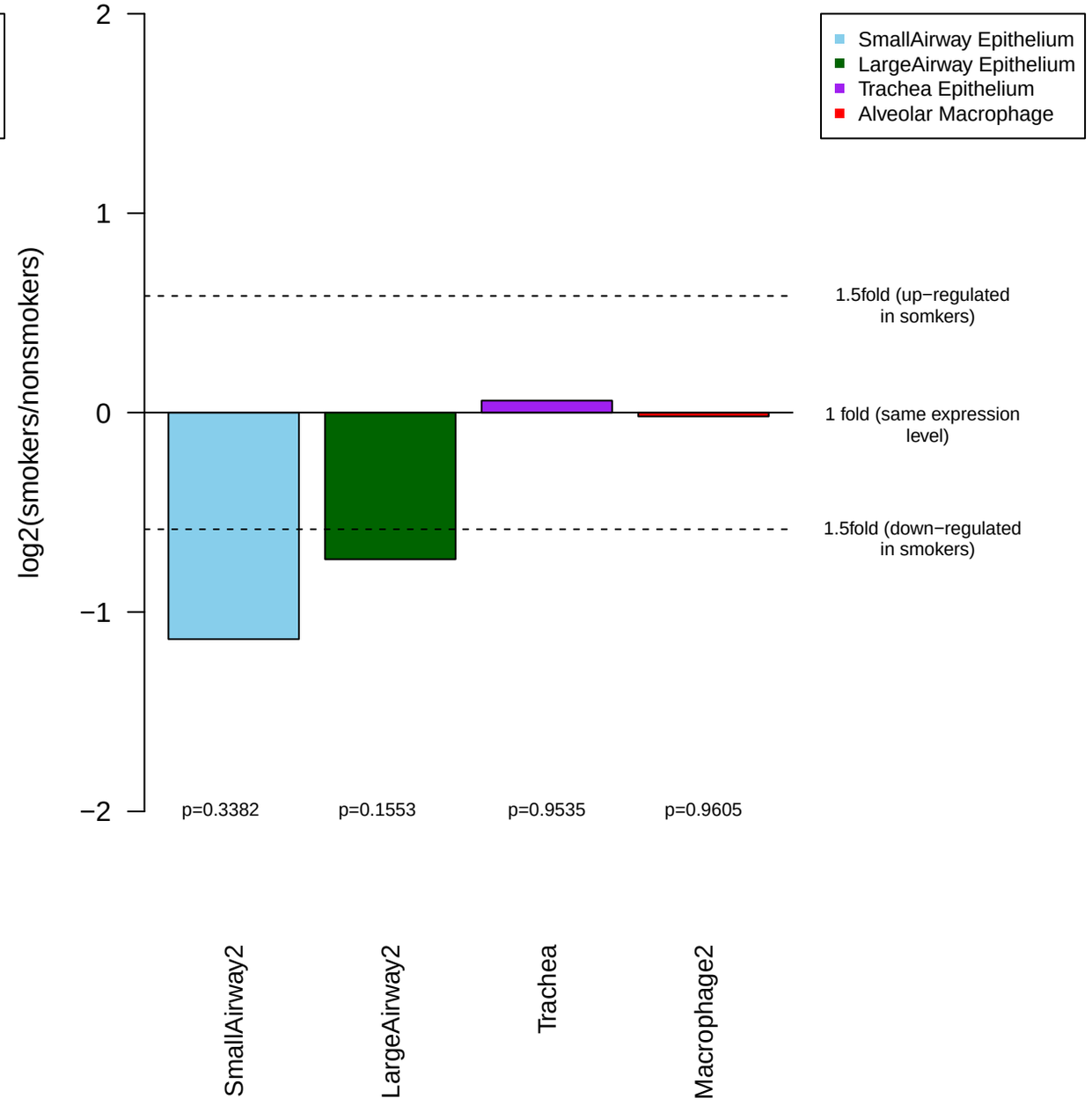
Gene expression ratio in Males

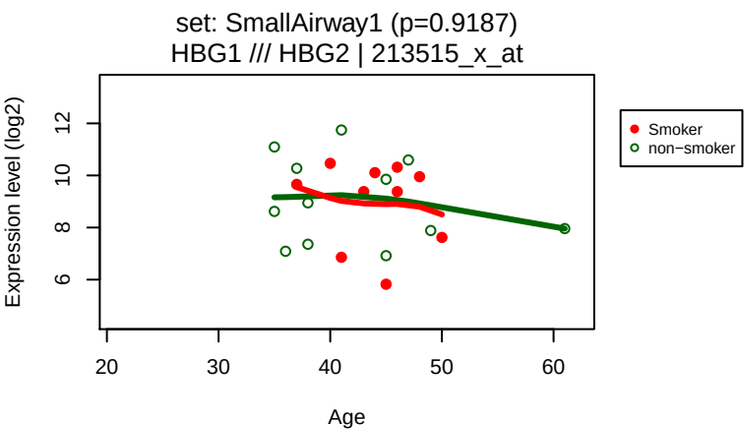
gene: HBG1 /// HBG2 | 213515_x_at

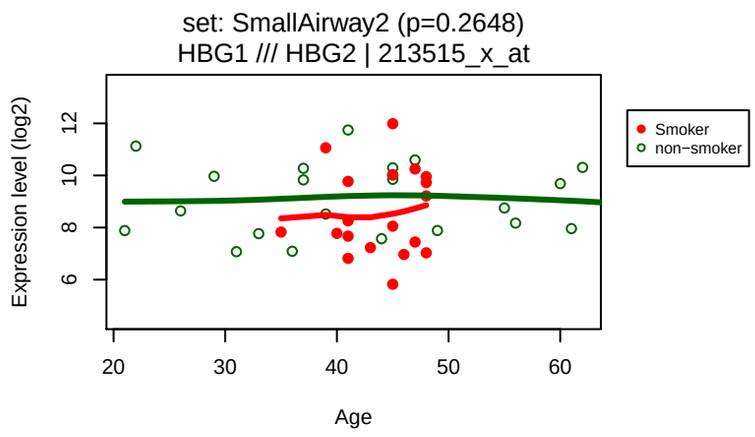


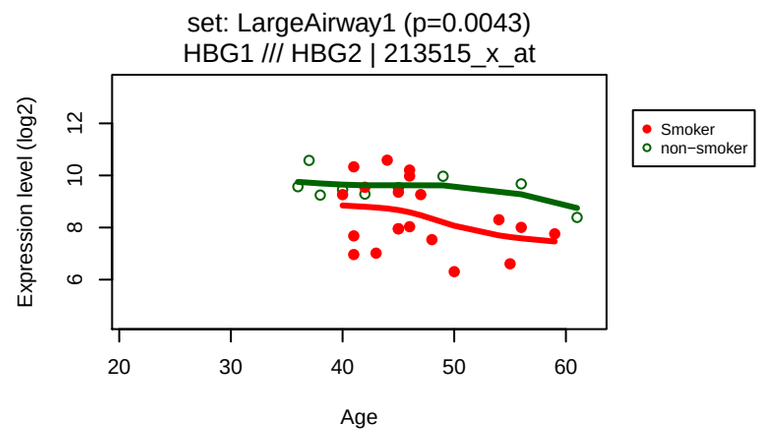
Gene expression ratio in Females

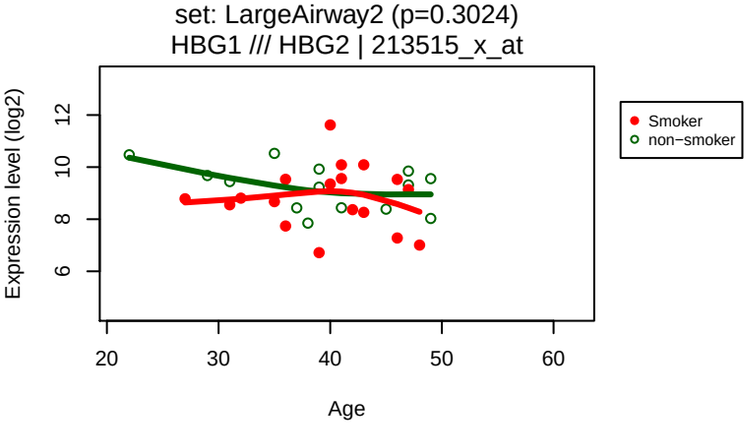
gene: HBG1 /// HBG2 | 213515_x_at

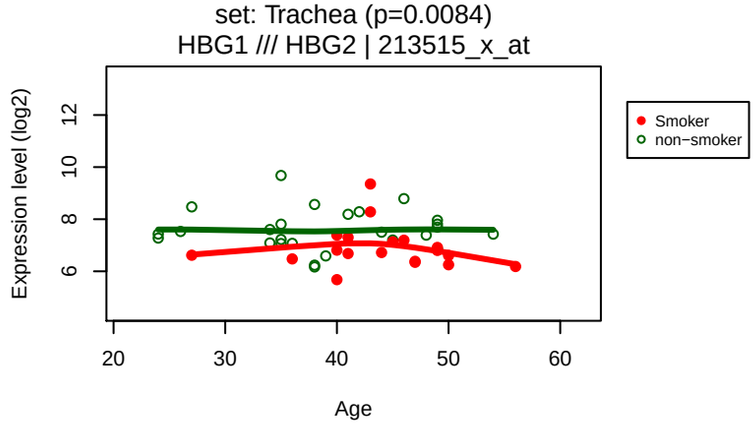


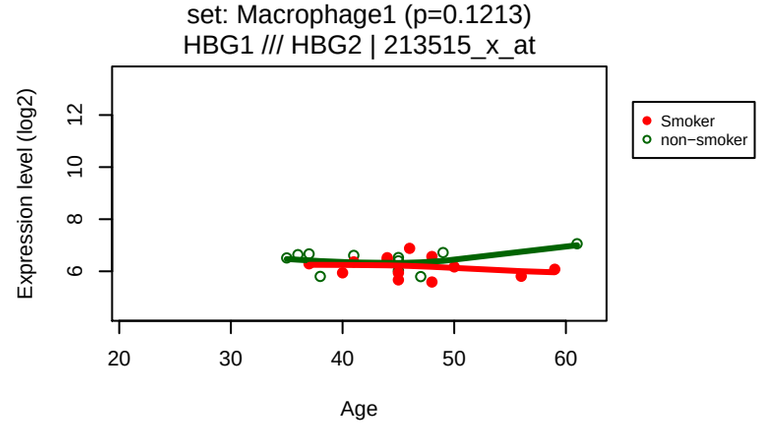

set: SmallAirway1 (p=0.9187)
HBG1 /// HBG2 | 213515_x_at
Expression level (log2)
Age
Smoker
non−smoker

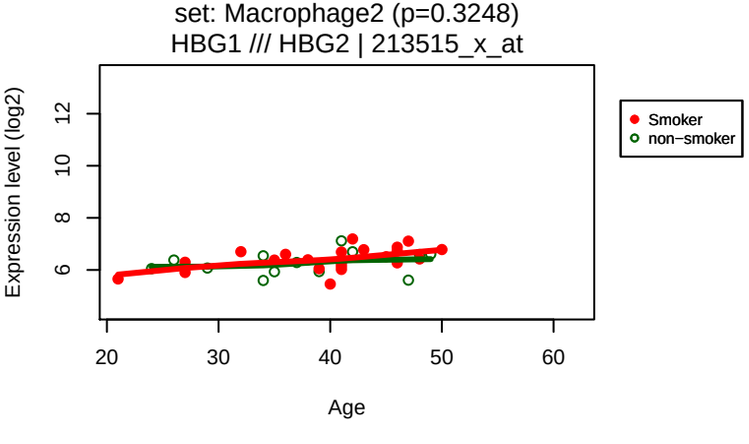

set: SmallAirway2 (p=0.2648)
HBG1 /// HBG2 | 213515_x_at
Expression level (log2)
Age
Smoker
non−smoker

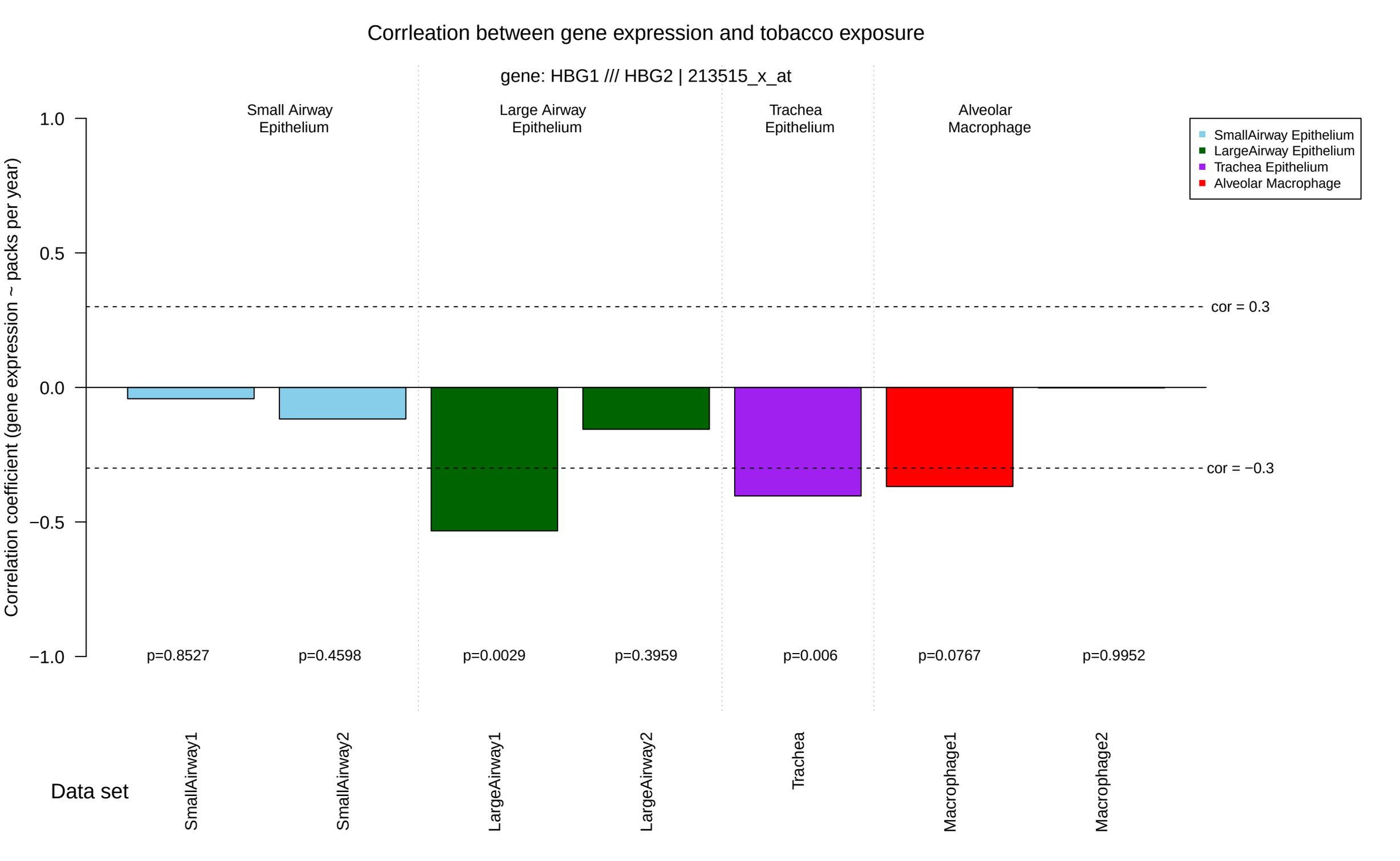

set: LargeAirway1 (p=0.0043)
HBG1 /// HBG2 | 213515_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3024)
HBG1 /// HBG2 | 213515_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0084)
HBG1 /// HBG2 | 213515_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1213)
HBG1 /// HBG2 | 213515_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3248)
HBG1 /// HBG2 | 213515_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HBG1 /// HBG2 | 213515_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8527
p=0.4598
p=0.0029
p=0.3959
p=0.006
p=0.0767
p=0.9952
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2